

Pythagoras: a universe ruled by numbers

Pythagoras was born on the island of Samos off the Ionian coast, but unlike the other Ionian philosophers he apparently did not stay there for long. He is believed to have travelled around the Mediterranean, until, aged about 40, he set up a pseudo-religious sect in the Greek colony of Croton in southern Italy. This eccentric genius had a particular talent for mathematics, making connections between geometry and arithmetic through the idea of squared and cubed numbers, and recognizing mathematical ratios inherent in music and acoustics. He also brought this knowledge to bear on philosophical questions, arguing that there was a structure to the cosmos (a word he coined) based on mathematical laws. The positions and movements of the heavenly bodies, he suggested, were analogous to the ratios of musical harmony (the so-called 'harmony of the spheres'). Pythagoras was the first to connect mathematics and philosophy, and the founder of a distinguished line of philosophers, including Descartes, Leibniz and Russell, who were also great mathematicians.



Pythagoras is often credited with discovering the mathematical equations relating to musical harmonies, supposedly after hearing the different notes emitted when hammers of different weights struck an anvil.